

## MIS and E-Business

BCA — Semester 5 — 2020 — Answer Sheet

Subject Code: CACS301

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### Group B

Attempt any SIX questions. [6×5=30]

2. How properties like ubiquity, richness and information density make e-commerce better than traditional commerce?

[5]

**Answer: Ubiquity:** E-commerce is available everywhere, at all times. Customers can shop from anywhere with internet access, 24/7. Traditional commerce is limited by physical store locations and operating hours. Ubiquity expands market reach beyond geographic boundaries and eliminates time constraints. **Richness:** E-commerce can deliver rich, interactive content including videos, 3D models, reviews, and personalized recommendations. Traditional commerce is limited to in-person demonstrations and static displays. Online platforms can provide detailed product information, user guides, and multimedia experiences that enhance customer understanding. **Information Density:** E-commerce enables unprecedented information density at low cost. Sellers can track detailed customer behavior, preferences, and purchase history. Prices can be adjusted dynamically based on demand, competition, and customer segments. Traditional commerce has limited ability to collect and analyze customer data. Online merchants can use data analytics to personalize offers, optimize inventory, and improve customer experience. **Combined Impact:** These properties together create a more efficient, personalized, and accessible marketplace that reduces transaction costs, eliminates intermediaries, and provides greater value to both buyers and sellers.

### 3. Differentiate B2C, B2B and C2C e-commerce with examples.

[5]

**Answer: B2C (Business-to-Consumer):** Transactions between businesses and individual consumers. Businesses sell products or services directly to end users. • Examples: Amazon, Flipkart, Daraz, Netflix • Characteristics: Large customer base, lower transaction volume per customer, marketing focused on individual preferences, payment via credit cards/digital wallets • Focus: Customer experience, brand loyalty, convenient checkout

**B2B (Business-to-Business):** Transactions between two businesses. One business sells products or services to another business. • Examples: Alibaba, IndiaMART, Salesforce, SAP • Characteristics: Smaller customer base, higher transaction values, longer sales cycles, negotiated contracts, bulk orders • Focus: Relationship management, supply chain efficiency, integration with buyer's systems

**C2C (Consumer-to-Consumer):** Transactions between individual consumers, often facilitated by a third-party platform. • Examples: eBay, Etsy, Facebook Marketplace, Hamrobazar • Characteristics: Peer-to-peer transactions, platform takes commission, user-generated listings, trust and rating systems • Focus: Community building, trust mechanisms, ease of listing/selling

**Comparison Table:**

| Aspect       | B2C                 | B2B                 | C2C                 |
|--------------|---------------------|---------------------|---------------------|
| Participants | Business + Consumer | Business + Business | Consumer + Consumer |
| Order Size   | Small               | Large               | Bulk                |
| Variable     |                     |                     |                     |
| Decision     |                     |                     |                     |
| Time         | Quick               | Long                | Complex             |
| Moderate     |                     |                     |                     |
| Payment      | Instant             | Credit/Invoice      | Platform-mediated   |
| Relationship | Transactional       | Strategic           | Community-based     |

### 4. What is the role of information superhighway in e-commerce? List the components of information superhighway.

[2+3]

**Answer: Role of Information Superhighway in E-Commerce:** The information superhighway is a global network of interconnected communication systems (primarily the Internet) that enables the rapid transmission of digital information across geographic boundaries. • **Global Connectivity:** Connects buyers and sellers worldwide, enabling global e-commerce markets • **Real-time Communication:** Supports instant messaging, video conferencing, and live customer support • **Data Transmission:** Enables secure transfer of payment information, order details, and digital products • **Marketing Channel:** Provides platforms for digital advertising, social media marketing, and search engine optimization • **Supply Chain Integration:** Connects suppliers, manufacturers, distributors, and retailers for efficient operations • **Infrastructure Foundation:** Serves as the backbone for all e-commerce activities including web hosting, cloud services, and content delivery networks

**Components of Information Superhighway:**

1. **Physical Infrastructure:** Fiber optic cables, satellites, cellular towers, routers, switches, and data centers
2. **Network Protocols:** TCP/IP, HTTP/HTTPS, FTP, and other communication standards
3. **Hardware Devices:** Computers, smartphones, tablets, servers, and networking equipment
4. **Software Applications:** Web browsers, email clients, e-commerce platforms, and mobile apps
5. **Content Providers:** Websites, streaming services, digital libraries, and online databases
6. **Service Providers:** ISPs (Internet Service Providers), cloud service providers, and hosting companies
7. **Human Users:** Consumers, businesses, governments, and educators who create and consume content

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**5. What is international information system? Describe the concepts of outsourcing and offshoring in the system.**

[1+4]

**Answer: International Information System:** An international information system is an information system that supports business operations across multiple countries. It handles diverse languages, currencies, regulations, time zones, and cultural differences. Such systems are essential for multinational corporations to coordinate global operations, manage international supply chains, and serve customers in different regions.

**Outsourcing:** Outsourcing is the practice of contracting specific business functions or IT services to external third-party providers, regardless of location. • **Key Points:** - Can be domestic or international - Focuses on delegating specific tasks (e.g., customer support, software development, data entry) - Company retains control over core business operations - Reduces operational costs and accesses specialized expertise • **Example:** A US company hiring an Indian firm to handle technical support

**Offshoring:** Offshoring is the relocation of business processes or operations to another country, typically to reduce costs or access new markets. • **Key Points:** - Always involves a different country - Can involve the company's own subsidiary (captive offshoring) or third-party providers - Often motivated by lower labor costs, tax incentives, or access to skilled talent - May involve manufacturing, service delivery, or IT operations • **Example:** Apple manufacturing iPhones in China; Google operating R&D centers in India

**Comparison:**

| Aspect    | Outsourcing                | Offshoring                           |
|-----------|----------------------------|--------------------------------------|
| Location  | Can be domestic or foreign | Always foreign                       |
| Ownership | Third-party provider       | Can be own subsidiary or third-party |
| Focus     | Specific function/task     | Entire process/operation             |
| Control   | Lower (external vendor)    | Higher (if captive)                  |

**6. Define availability, authentication and authorization. How they can be enforced in e-commerce security?**

[3+2]

**Answer: Availability:** Availability ensures that e-commerce systems and data are accessible to authorized users whenever needed. It guarantees that websites, payment gateways, and databases remain operational and responsive. • **Enforcement:** Redundant servers, load balancing, DDoS protection, backup systems, disaster recovery plans, SLA monitoring, and cloud-based auto-scaling

**Authentication:** Authentication is the process of verifying the identity of a user, device, or system. It ensures that the entity requesting access is who it claims to be. • **Methods:** Passwords, multi-factor authentication (MFA), biometric verification (fingerprint, facial recognition), digital certificates, token-based authentication, and single sign-on (SSO) • **Enforcement:** Strong password policies, MFA implementation, CAPTCHA for bot detection, session management, and secure login protocols (OAuth, SAML)

**Authorization:** Authorization determines what resources and actions an authenticated user is permitted to access. It defines the permissions and privileges assigned to users. • **Enforcement:** Role-Based Access Control (RBAC), Access Control Lists (ACL), permission matrices, principle of least privilege, and regular access audits

**How They Work Together in E-Commerce:**

- 1. Authentication First:** Customer logs in with username/password + OTP (MFA)
- 2. Authorization Next:** System checks what the customer can access (view orders, edit profile, but not admin functions)
- 3. Availability Ensured:** Website remains online during peak shopping periods (e.g., Black Friday) through load balancing and CDN

**Security Defense:** Together with encryption (confidentiality) and integrity checks, these form the core security framework protecting e-commerce transactions and customer data.

## 7. What is e-checking? Describe its working mechanism.

[2+3]

**Answer: E-Checking (Electronic Checking):** E-checking is the electronic version of a paper check used for online payments. It contains the same information as a paper check (account number, routing number, payment amount) but is transmitted electronically. E-checks are processed through the Automated Clearing House (ACH) network and are governed by the same laws as paper checks. **Working Mechanism of E-Checking:** 1. **Authorization:** The payer authorizes the payment by providing bank account details (account number, routing number) and signing an authorization form (digital or physical). 2. **Payment Setup:** The merchant or payment processor creates an electronic check with: - Payer's bank account number - Bank routing number - Payment amount - Date and authorization 3. **Encryption & Security:** The e-check is encrypted and digitally signed to ensure security and non-repudiation. 4. **Transmission:** The encrypted e-check is sent through the ACH network from the payer's bank to the payee's bank. 5. **Verification:** The payer's bank verifies that sufficient funds are available and that the account is valid. 6. **Clearing & Settlement:** Funds are debited from the payer's account and credited to the payee's account. This typically takes 1-3 business days. 7. **Confirmation:** Both parties receive confirmation of the transaction. **Advantages:** Lower processing fees than credit cards, no need for physical handling, faster than traditional mail, automatic record keeping **Disadvantages:** Slower than credit card processing, requires bank account information, limited consumer protection, not suitable for instant purchases

## 8. Describe the website design criteria that one should show while designing an e-commerce website.

[5]

**Answer: Website Design Criteria for E-Commerce:** 1. **User-Friendly Navigation:** - Clear menu structure with logical categories - Search functionality with filters and autocomplete - Breadcrumb navigation to show user's location - Consistent layout across all pages 2. **Responsive Design:** - Website must work seamlessly on desktop, tablet, and mobile devices - Touch-friendly buttons and forms - Fast loading times on all screen sizes - Mobile-first approach for growing mobile commerce 3. **Fast Loading Speed:** - Optimize images and use content delivery networks (CDN) - Minimize HTTP requests and enable browser caching - Use lazy loading for images and videos - Target page load time under 3 seconds 4. **Secure Payment Integration:** - SSL certificates for HTTPS encryption - Multiple payment options (credit cards, digital wallets, COD) - PCI DSS compliance for card data security - Clear display of security badges and trust seals 5. **High-Quality Product Presentation:** - Multiple product images with zoom functionality - Detailed product descriptions and specifications - Customer reviews and ratings - 360-degree views or product videos where applicable 6. **Streamlined Checkout Process:** - Minimal steps to complete purchase (ideally 3 steps or fewer) - Guest checkout option - Clear progress indicators - Auto-fill for returning customers 7. **Trust Signals:** - Visible contact information and customer support - Clear return and refund policies - Company information and physical address - Social proof (testimonials, user counts, media mentions) 8. **SEO Optimization:** - Search engine friendly URLs and meta tags - Structured data for products - Fast mobile performance (Core Web Vitals) - Quality content and blog integration

## Group C

Attempt any TWO questions. [2×10=20]

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**9. What do you mean by Wireless Application Protocol (WAP)? How it works? Discuss the layered architecture of WAP.**

[2+2+6]

**Answer: Wireless Application Protocol (WAP):** WAP is an open international standard for applications that use wireless communication. It enables mobile devices (phones, PDAs) to access internet content and services. WAP was designed to work with low-bandwidth, high-latency wireless networks and devices with small screens and limited input capabilities. **How WAP Works:** 1. **Request:** User enters a URL or clicks a link on a WAP-enabled mobile device 2. **WAP Gateway:** The request is sent to a WAP gateway, which acts as a proxy between the mobile network and the internet 3. **Protocol Translation:** The gateway converts WAP protocols to standard HTTP/TCP and fetches the requested content from the web server 4. **Content Encoding:** The gateway compresses and encodes the content into WML (Wireless Markup Language) or XHTML-MP 5. **Response:** The encoded content is sent back to the mobile device through the wireless network 6. **Rendering:** The device's WAP browser (microbrowser) renders the content on the small screen

**WAP Layered Architecture:**

| Layer Name | Function                | Equivalent in Web                                                                                 |                          |
|------------|-------------------------|---------------------------------------------------------------------------------------------------|--------------------------|
| Layer 5    | Application Layer (WAE) | Provides environment for wireless applications; includes WML, WMLScript, and telephony APIs       | HTTP + HTML + JavaScript |
| Layer 4    | Session Layer (WSP)     | Manages sessions between client and server; supports connection-oriented and connectionless modes | HTTP/HTTPS               |
| Layer 3    | Transaction Layer (WTP) | Provides reliable request-response transactions with minimal overhead                             | TCP/UDP                  |
| Layer 2    | Security Layer (WTLS)   | Provides encryption, data integrity, and authentication; equivalent to TLS/SSL                    | TLS/SSL                  |
| Layer 1    | Transport Layer (WDP)   | Provides datagram service, abstracting underlying network (GSM, CDMA, SMS)                        | UDP                      |
| Layer 0    | Bearer Networks         | Physical network: GSM, GPRS, CDMA, SMS, USSD                                                      | Ethernet, Wi-Fi          |

**Key Features:** Optimized for low bandwidth, handles intermittent connectivity, supports push services, device-independent content adaptation, and security through WTLS.

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10. How important web contents are for e-commerce systems? Discuss their types. Justify, with examples, how web contents can be used to promote cross-selling, up-selling and promotions in e-commerce.

[2+2+6]

**Answer: Importance of Web Content in E-Commerce:** Web content is the foundation of e-commerce success. It drives traffic, engages customers, builds trust, and converts visitors into buyers. • **SEO:** Quality content improves search engine rankings and organic traffic • **Customer Education:** Informs customers about products, features, and benefits • **Trust Building:** Reviews, testimonials, and detailed descriptions reduce purchase anxiety • **Brand Identity:** Content communicates brand values and personality • **Conversion:** Compelling product descriptions and images increase purchase likelihood • **Customer Support:** FAQ pages, guides, and tutorials reduce support costs

**Types of Web Content:** 1. **Product Content:** Descriptions, specifications, images, videos, 360° views 2. **Marketing Content:** Blog posts, articles, social media posts, email newsletters 3. **Transactional Content:** Pricing, shipping info, return policies, terms of service 4. **User-Generated Content:** Reviews, ratings, Q&A, photos from customers 5. **Interactive Content:** Quizzes, configurators, calculators, chatbots 6. **Visual Content:** Infographics, banners, product photos, demo videos 7. **Educational Content:** How-to guides, tutorials, comparison articles

**Cross-Selling with Web Content:** Cross-selling recommends complementary products to increase order value. • **Example:** Amazon's "Frequently Bought Together" section. When viewing a laptop, the page shows laptop bag, mouse, and keyboard. Content includes product images, compatibility info, and bundle pricing. • **Technique:** "Complete the Look" content on fashion sites shows matching accessories

**Up-Selling with Web Content:** Up-selling encourages customers to purchase a higher-end version of a product. • **Example:** Apple's iPhone comparison table showing features of different models side-by-side, highlighting the benefits of the Pro model (better camera, longer battery). Content includes detailed spec comparisons and benefit-driven descriptions. • **Technique:** "Upgrade and Save" messaging with feature comparison charts

**Promotions with Web Content:** Promotional content drives urgency and increases conversion rates. • **Example:** Flash sale countdown timers, "Limited Stock" badges, seasonal campaign banners. Daraz's "11.11 Sale" uses dedicated landing pages with countdown timers, discount badges, and curated product collections. • **Technique:** Personalized email content with exclusive discount codes based on browsing history

**Combined Strategy:** Netflix uses content to promote cross-selling ("Because you watched X, you might like Y") and up-selling (premium plan benefits highlighted in comparison tables) through personalized web content.

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**11. Discuss in detail the security defense strategies that you can implement while securing e-commerce systems from security attacks and threats.**

[10]

**Answer: Security Defense Strategies for E-Commerce:**

- 1. Encryption and SSL/TLS:** • Implement SSL/TLS certificates to encrypt all data transmitted between customers and servers • Use HTTPS for all pages, not just checkout • Encrypt sensitive data at rest (customer information, payment details) • Implement end-to-end encryption for messaging and communications
- 2. Firewalls and Intrusion Detection/Prevention:** • Deploy Web Application Firewalls (WAF) to filter malicious traffic • Use network firewalls to control incoming and outgoing traffic • Implement IDS/IPS to detect and block suspicious activities in real-time • Regularly update firewall rules based on threat intelligence
- 3. Secure Payment Processing:** • Comply with PCI DSS (Payment Card Industry Data Security Standard) • Use tokenization to replace card numbers with non-sensitive tokens • Implement 3D Secure authentication for card payments • Consider using trusted payment gateways (Stripe, PayPal) rather than storing card data
- 4. Authentication and Access Control:** • Enforce strong password policies (minimum length, complexity requirements) • Implement Multi-Factor Authentication (MFA) for admin and customer accounts • Use Role-Based Access Control (RBAC) for internal systems • Implement account lockout policies after failed login attempts • Use OAuth/SAML for secure third-party authentication
- 5. Regular Security Audits and Testing:** • Conduct penetration testing to identify vulnerabilities • Perform code reviews and security scans • Use automated vulnerability scanning tools • Engage third-party security auditors annually • Implement bug bounty programs
- 6. Data Backup and Disaster Recovery:** • Maintain regular backups of all critical data • Store backups in geographically separate locations • Test disaster recovery procedures regularly • Implement point-in-time recovery capabilities • Use redundant servers and failover mechanisms
- 7. DDoS Protection:** • Use DDoS mitigation services (Cloudflare, AWS Shield) • Implement rate limiting on APIs and login endpoints • Use Content Delivery Networks (CDN) to absorb traffic spikes • Monitor traffic patterns for anomalies
- 8. Software Updates and Patch Management:** • Keep all software, frameworks, and libraries up to date • Apply security patches promptly • Remove unused plugins and dependencies • Use automated patch management tools
- 9. Employee Training and Awareness:** • Train staff on phishing recognition and social engineering • Establish clear security policies and procedures • Conduct regular security awareness simulations • Limit access to sensitive systems based on job roles
- 10. Legal and Compliance Measures:** • Implement privacy policies compliant with GDPR, CCPA • Maintain audit logs of all transactions • Use digital signatures for non-repudiation • Comply with consumer protection laws • Implement data retention and deletion policies

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*Note: These answers are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.*